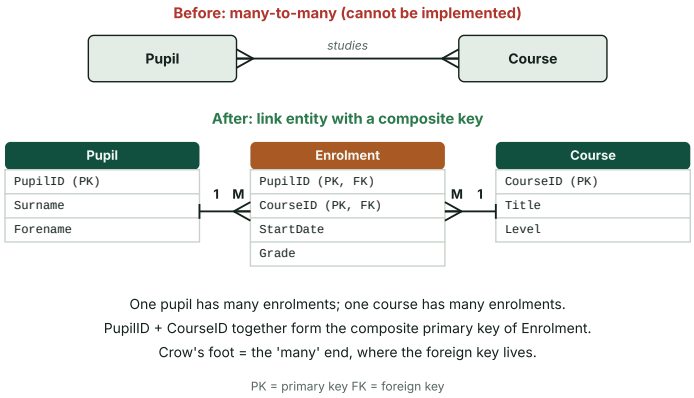
Entity-relationship modelling

IS02 FACT FILE

Specification: students should be able to

- Describe entities, attributes and relationships
- Explain primary, foreign, composite and candidate keys
- Draw and interpret entity-relationship (ER) diagrams showing 1:1, 1:M and M:N relationships
- Resolve many-to-many relationships using a link entity

Key words: entity, attribute, relationship, primary key, foreign key, composite key, candidate key, one-to-many, many-to-many, link entity, crow's foot, degree



A many-to-many relationship between Pupil and Course resolved by an Enrolment link entity with a composite key, giving two one-to-many relationships

Entities, attributes and keys

An entity is any object, person, place or event about which data is held; its attributes are the data items describing it. Every entity needs a primary key that uniquely identifies each occurrence. A foreign key is an attribute in one table that is the primary key of another, creating the link between them. Where no single attribute is unique, a composite key made from two or more attributes is used. Any attribute that could be chosen as the primary key is a candidate key.

Relationships

The degree of a relationship describes how many occurrences of one entity relate to another: one-to-one, one-to-many or many-to-many. In crow's foot notation the many end is drawn with three prongs. In a one-to-many relationship the foreign key goes in the table at the many end. A one-to-one relationship can often be merged into a single table unless the data is optional or needs separate security.

Resolving many-to-many

A many-to-many relationship cannot be implemented directly because a field cannot hold a list of values. It is resolved by introducing a link entity whose primary key is usually a composite of the two original primary keys. The link entity turns one M:N relationship into two 1:M relationships and is often where useful attributes live, such as the date of an enrolment or the grade achieved.

T Key terms from the mark schemes

Entity
An object, person, place or event about which data is stored; it becomes a table in a relational database.

Attribute
A property or characteristic of an entity; it becomes a field in a table.

Primary key
An attribute, or combination of attributes, that uniquely identifies each record in a table.

Foreign key
An attribute in one table that is the primary key of another table and is used to link the two tables.

Composite key
A primary key made up of two or more attributes which together uniquely identify a record.

Link entity
An entity introduced to resolve a many-to-many relationship into two one-to-many relationships.

Exam focus

You may be asked to draw an ER diagram from a scenario or complete one. Marks are given for each correct entity, each correct relationship with the crow's foot at the correct end, and for resolving M:N with a correctly named link entity. When asked to 'explain why' a link entity is needed, state that an M:N relationship cannot be implemented in a relational database and that the link entity creates two 1:M relationships. Underline primary keys and use an asterisk or the words 'foreign key' – be consistent.

? Check yourself

1. State what is meant by a composite key. [1]
2. Explain the purpose of a foreign key. [2]
3. A library allows each member to borrow many books, and over time each book is borrowed by many members. Draw an ER diagram showing how this relationship would be implemented in a relational database. [4]

Answers: 1. A composite key is a primary key made up of two or more attributes which together uniquely identify each record, e.g. (PupilID, CourseID) in Enrolment (1). 2. A foreign key is an attribute in one table that is the primary key of another table (1) / it is used to create the link (relationship) between the two tables, so related records can be joined, and allows referential integrity to be enforced (1). 3. Three entities drawn: Member, Book and a link entity, e.g. Loan (1) / Member to Loan is one-to-many with the crow's foot at Loan (1) / Book to Loan is one-to-many with the crow's foot at Loan (1) / no direct many-to-many line remains between Member and Book, i.e. M:N correctly resolved (1). Diagram: Member 1 ——< Loan >—— 1 Book. Accept any sensible name for the link entity (Borrowing, Issue).